Distinguishing biological transport from passive Brownian diffusion

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Abstract

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Since Aristotle, movement has been a primary metric for defining life [1]; however, modern biophysics dictates that not all microscopic motion indicates biological viability. Active biological transport requires the continuous expenditure of metabolic energy to generate directed movement, serving as a fundamental physical signature of living systems [4]. By contrast, Brownian motion—the random, thermally driven displacement of microscopic particles—is a defining feature of passive matter at thermal equilibrium. To establish a quantitative baseline for passive motion, we first characterized Brownian diffusion by tracking polystyrene microspheres (1–5 µm) in fluids spanning a 12-fold viscosity range (0.32–3.7 mPa s). This systematic study validated the Stokes-Einstein relation to within 7 and confirmed a mean-squared-displacement (MSD) exponent $\alpha = 0.92 \pm 0.15$, consistent with classical thermal equilibrium. We then applied the identical tracking

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INTRODUCTION

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A polystyrene bead suspended in water moves, but it does not choose where to go. Its trajectory is a random walk driven entirely by thermal collisions with the surrounding solvent molecules, and its statistical properties are predicted exactly by Einstein's 1905 theory of Brownian motion [6]. no preferred direction, no temporal correlation, and no purpose: it is the kinematic fingerprint of matter at thermal equilibrium.

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THEORY

The diffusion coefficient D is a crucial quantity that represents how rapidly a particle moves due to thermal agitation within a fluid medium. A particle performing a "random walk" takes N independent steps of characteristic length b in random directions; along any single coordinate axis, the mean displacement vanishes, but the mean square displacement grows linearly: $\langle x^2 \rangle = Nb^2$ [8]. Since the x and y displacements are independent, the total two-dimensional MSD is $\langle \Delta r^2 \rangle = \langle \Delta x^2 \rangle + \langle \Delta y^2 \rangle$. In the continuum regime, Einstein shows

that for a freely diffusing particle tracked in two dimensions, this grows linearly with lag time τ [6]:

$$\langle \Delta r^2(\tau) \rangle = 4D\tau \,. \tag{1}$$

The diffusion coefficient itself is determined from the Stokes-Einstein relation. For a sphere of radius r suspended in a viscosity η fluid at absolute temperature T, thermal agitation counters drag from the fluid's viscosity and gives [6]

$$D = \frac{k_B T}{6\pi\eta r} \,. \tag{2}$$

Here $k_B = 1.381 \times 10^{-23}$ J/K is the fundamental Boltzmann's constant. The two important relations in the above equation are $D \propto 1/\eta$ at fixed bead size, and $D \propto 1/r$ at fixed viscosity. These provide theoretical independent properties expected from Brownian motion. This experiment tests both by varying the bead diameter (1–5 µm) and the fluid viscosity (0.32–3.7 mPa s).

The D value is measured from trajectories using three independent statistical methods: variance of frame-to-frame displacements, Gaussian fit to the displacement histogram, and finally the slope of the MSD curve. All three are required to agree as an internal consistency check [5]. Generally, the MSD may not grow linearly for large lag times τ. The standard generalization replaces the linear dependence with a power law [5]:

$$\langle \Delta r^2(\tau) \rangle = 4D\tau^{\alpha} \,. \tag{3}$$

The exponent α classifies the type of motion: $\alpha = 1$ for normal Brownian diffusion [9]. A measured α significantly greater than 1 is clear evidence that the system is not at equilibrium; i.e., work is being put into the system to create a directed motion. In a living cell, this energy comes from ATP compounds, powering molecular motors along the cytoskeletal filaments, cell wall and towards the nucleus [3, 4]; in contrast, no such mechanism is present for an inert bead, forcing the α below 1.

The motion of active biological systems can be contrasted quantitatively with the motion of passive beads of comparable size [2]. This comparison is made precise by using the α exponent: bead data establish the $\alpha \approx 1$ baseline expected at thermal equilibrium, and any significant excess, more than 1 σ in the onion (*Allium cepa*), the data shows evidence of active, energy-consuming transport and hence a pivotal signature of organic life.

Further, several uncertainty and instrumental corrections are considered when comparing measured and predicted D values. As a bead of radius R travels between two parallel glass surfaces, hydrodynamic coupling to both walls reduce the diffusion coefficient. Approximating the total wall effect as a superposition of two independent single-wall Faxén corrections [12] gives

$$D_{\parallel} = D_0\big(1 - \tfrac{9}{16}\,s + \tfrac{1}{8}\,s^3 - \cdots\big)\,, \quad s = \frac{R}{h} + \frac{R}{L-h}\,, \tag{4}$$

where h is the height of the bead centre above the lower glass surface and D_0 is the bulk Stokes-Einstein value from Eq. (2). For a chamber depth $L = 82.5$ µm, this correction reduces D by 1–7% at the midplane and substantially more near the walls. Since the exact position on the z-plane cannot be determined, the expectation would be a range of values spanning the midpoint to the wall. The viscosity of water-glycerol mixtures is calculated using the empirical model of Cheng [13], giving $\eta = 0.98$, 1.69, and 3.59 mPa s for pure water, 20%, and 40% glycerol by volume at 22 °C. Pure acetone ($\eta = 0.32$ mPa s) is used as a low-viscosity solvent to extend the range further.

Crucially, every one of these corrections acts to *reduce* α; none can produce superdiffusion, i.e., make particles diffuse faster. Any measured α significantly above 1 is therefore not a sign of the instruments interfering, but indicates a non-equilibrium process like drift in the sample.

METHODS

Apparatus

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All data were collected on the same instrument: an Olympus BX51 upright bright-field microscope with a 100× oil-immersion objective (NA 1.25), equipped with a FLIR BlackFly U3-13Y3M monochrome camera (1440 × 1080 px) recording at 29 fps via National Instruments Vision Assistant. The sample chamber was a glass slide and No. 1 coverslip separated by adhesive tape spacers ($L = 82.5 \pm 2.5$ µm). ...

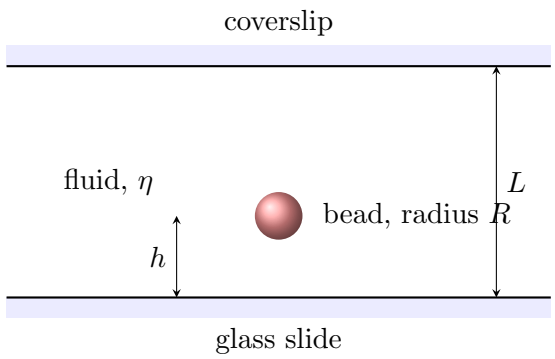

FIG. 1. Cross-section of the sample chamber (not to scale). A bead of radius R diffuses in fluid of viscosity η between glass surfaces separated by $L = 82.5$ µm. The height h determines the Faxén wall correction [Eq. (4)].

Camera calibration and noise floor

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Bead sample preparation

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Onion cell preparation

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Tracking and analysis

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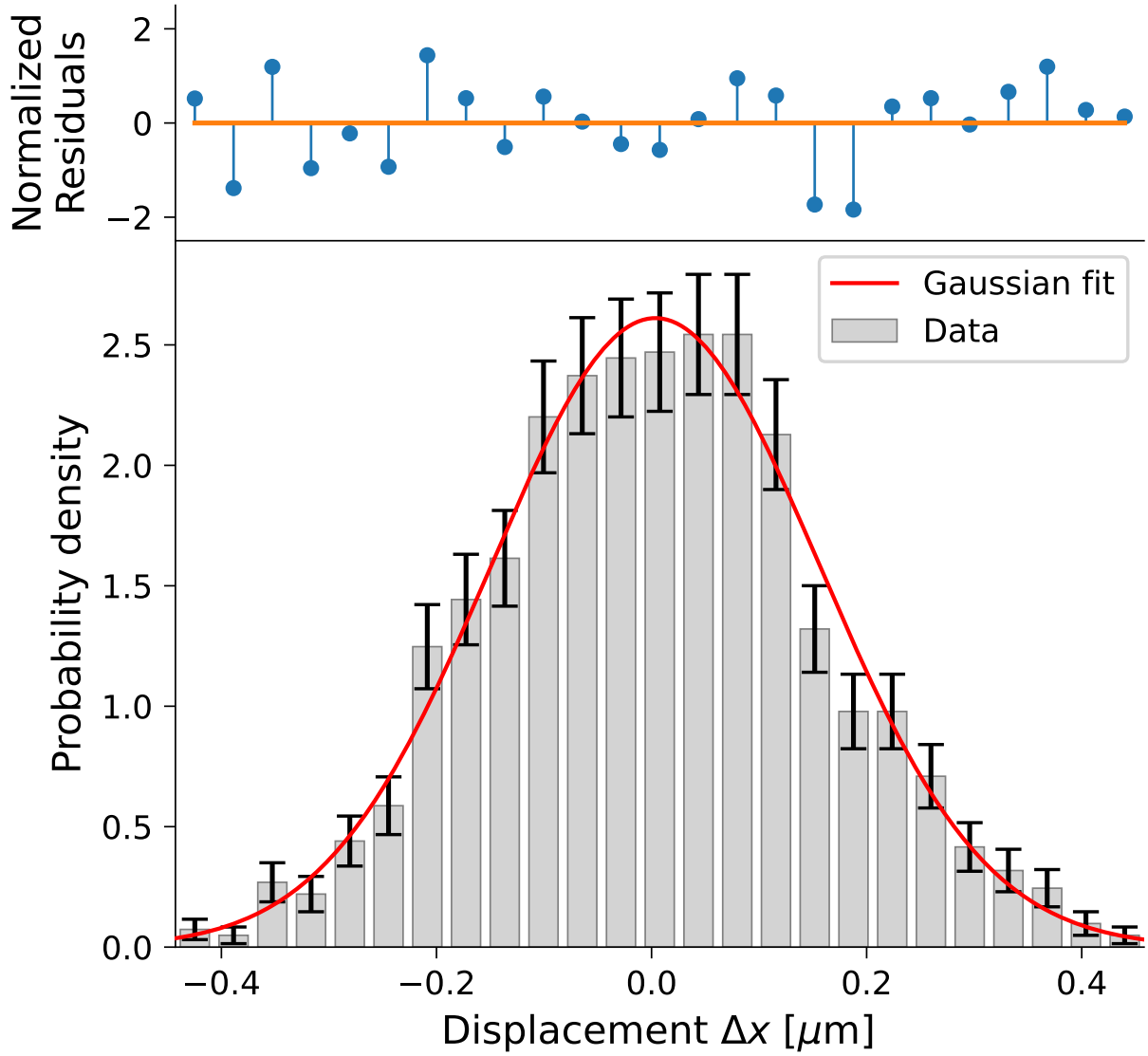

FIG. 2. $\text{gaussian}_d isplacement_f it - 026 - 03 - 03r14 - 2_1 mu - 0_5 p - 4ul - acetone - 498ul - trial2$

Comparison with theory

The MSD exponent of dead matter

DISCUSSION

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Bead (µm)	Solvent	N	$D_{\mathrm{var}} \pm \sigma$	D_{gauss}	D_{MSD}	D_{theory}	Ratio	α
1.0	Water	1	0.407 ± 0.009	0.404	0.382	0.433	0.94	0.81
1.0	20% Gly	2	0.262 ± 0.010	0.259	0.237	0.251	1.04	1.08
2.1	Water	1	0.071 ± 0.025	0.063	0.073	0.203	0.35	1.23
2.1	20% Gly	2	0.123 ± 0.018	0.121	0.122	0.117	1.05	0.79
2.1	40% Gly	2	0.023 ± 0.017	0.020	0.023	0.056	0.42	0.91
2.1	Acetone	2	0.173 ± 0.245	0.171	0.362	0.623	0.28	0.99
3.0	Water	3	0.099 ± 0.029	0.093	0.069	0.140	0.71	0.68
3.0	20% Gly	3	0.087 ± 0.022	0.086	0.077	0.081	1.07	0.99
3.0	40% Gly	2	0.006 ± 0.003	0.003	0.002	0.038	0.15	0.90
3.0	Acetone	1	0.196 ± 0.011	0.200	0.200	0.431	0.46	0.81
5.0	Water	2	0.009 ± 0.012	0.009	0.007	0.082	0.10	0.77
5.0	20% Gly	3	0.019 ± 0.017	0.021	0.020	0.047	0.42	1.14
5.0	40% Gly	4	<0.001	<0.001	<0.001	0.022	<0.02	1.07
5.0	Acetone	3	0.064 ± 0.008	0.066	0.064	0.251	0.25	0.86

TABLE I. Caption...

CONCLUSIONS

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ACKNOWLEDGMENTS

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